Step 1:

Research Questions

The authors investigate whether different coaching styles (positive-reinforcement vs. autonomous-supportive) and different viewpoints (first-person vs. third-person) influence balance performance, user experience, and engagement in a VR rehabilitation environment.

Hypotheses

The paper proposes four hypotheses: that positive-reinforcement will improve performance and user experience, and that the third-person viewpoint will also improve performance and user experience compared to first-person.

Sample Size Determination

They ran a G*Power calculation for their repeated-measures setup, and it showed that they only needed sixteen participants to have enough statistical power for the study.

Study Design

This was a 2 × 2 within-subjects design with two independent variables (coaching style and viewpoint) and multiple dependent variables measuring steadiness, foot height, mistakes, embodiment, usability, enjoyment, and task load.

Statistical Methods

The authors used two-way repeated-measures ANOVA to evaluate the effects of coaching style and viewpoint, followed by pairwise t-tests for significant findings.

Descriptive Analysis

The study includes descriptive statistics, such as means, standard deviations, and standard errors, which appear in Table 2 to summarize performance across conditions.

Group Comparisons

Because the design included four experimental conditions created by the combination of the two variables, the authors used ANOVA as the appropriate method to compare more than two groups at once.

Results Reporting

The results are reported clearly with F-values, p-values, effect sizes, and visual summaries that explain which main effects and interactions were significant.

Visualizations

The authors use box plots, bar charts, and NASA-TLX graphs, which make trends and group differences easier to interpret than tables alone.

Qualitative Analysis

The paper includes participant comments about tracking accuracy, comfort, the effectiveness of the virtual coach, viewpoint preferences, and learning effects across trials.

Hypothesis Discussion

The authors concluded that positive-reinforcement coaching improved steadiness and subjective experience, like supporting H1 and H2, while viewpoint did not produce the expected advantages, such as rejecting H3 and H4.

Limitations and Future Work

The authors acknowledge limitations such as Kinect tracking errors, issues with wearing glasses in VR, learning effects, and small sample size, and they propose future research including improved sensors, more feedback types, clinical testing, and real-world comparisons.